

The Edge of Innovation: Rethinking computing for AI in the Post-Moore's Era

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Abstract—The surge in AI applications demands efficient computing frameworks capable of processing vast datasets with minimal latency, prompting the emergence of edge computing. By bringing computation and data storage closer to where it's needed, edge computing reduces reliance on centralized data centers and mitigates bandwidth issues. However, its rapid evolution faces challenges like hardware limitations, energy consumption, and environmental impact, prompting a re-evaluation of Moore's Law. This case study examines the intricate dynamics between the advancing edge computing landscape and the enduring principles of Moore's Law. By emphasizing the transformative impact of technologies like Neural Processing Units (NPUs) and accelerated Graphics Processing Units (GPUs) in edge computing scenarios, the research advocates for a balanced approach to computational development that harmonizes efficiency with sustainability.

I. INTRODUCTION

In the recent era, characterized by the rapid proliferation of the Internet of Things (IoT), an increase in smart device usage, and the widespread adoption of wireless networking, the landscape of artificial intelligence (AI) applications has experienced substantial growth. This surge is largely attributed to the advancements and increasing reliance on edge computing, a paradigm that processes data closer to where it is generated, thus reducing latency and enhancing efficiency. According to the International Data Corporation (IDC), global spending on edge computing is projected to reach \$232 billion by 2024, signifying a 15.4% increase from the previous year, with expectations of continued growth to nearly \$350 billion by 2027 [1][2]. This investment underscores the vital role of edge computing in deploying AI applications, necessitating a distributed architectural approach to meet the scalability and performance demands of modern technology.

The convergence of edge computing with AI applications heralds a new frontier in computational technology, where the limitations of traditional centralized data processing are overcome by distributing computational tasks closer to data sources. This shift is not without its challenges, particularly in the realms of hardware capabilities, energy consumption, network usage, and environmental impact. Addressing these challenges requires a reevaluation of the computational frameworks underpinning current technologies, with a particular focus on the hardware processing capabilities of edge devices.

This paper delves into the current state of the art in edge computing from a hardware processing perspective, highlighting the critical roles of Neural Processing Units (NPUs), Tensor Processing Units (TPUs), and AI-accelerated Graph-

ics Processing Units (GPUs). These specialized hardware components are essential for navigating the computational bottlenecks inherent in edge devices, enabling them to support the complex algorithms that underlie contemporary AI applications. Furthermore, this study explores avenues for innovation and improvement in edge computing hardware, aiming to enhance the robustness and efficiency of these systems to better accommodate the growing demands of AI algorithms.

Through an examination of the latest advancements in NPUs, TPUs, and AI-accelerated GPUs, this research aims to illuminate the path toward a more efficient, sustainable, and capable edge computing paradigm. In doing so, it seeks to contribute to the ongoing discourse on the symbiotic relationship between edge computing and AI, advocating for a balanced approach to computational development that aligns with the exponential growth model of Moore's Law. By fostering a deeper understanding of the current capabilities and future potential of edge computing hardware, this paper endeavors to guide the industry toward realizing the full promise of AI applications in the digital age.

A. Background and Significance

Computing has undergone a significant shift from the era of centralized mainframe computing, characterized by terminals lacking processing capabilities, to today's landscape where smart devices independently process and respond to data. This historical perspective(which was depicted in the below Fig1) lays the groundwork for appreciating the significance of edge computing and its potential in contemporary technological applications.

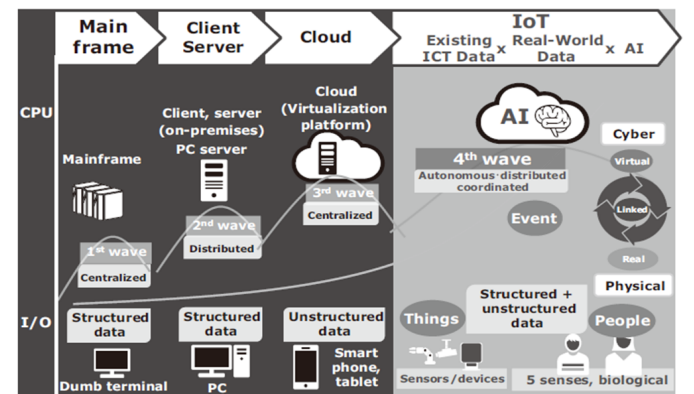


Fig. 1: TimeLine Phase of Computing

As we chart the course from structured data managed by traditional PCs to the intricate interplay of structured and unstructured data processed by today's smartphones and sensors, the role of artificial intelligence becomes ever more prominent. AI algorithms are no longer confined to the cloud but are increasingly deployed on edge devices—bringing intelligence closer to where the data originates and actions are performed.

With the advent of the Internet of Things, the proliferation of smart devices, and widespread wireless networking, the demand for edge computing has skyrocketed. IDC's projections reflect a growing recognition of its critical role, with significant investments anticipated to reach \$232 billion by 2024 and potentially nearing \$350 billion by 2027 [14][15]. This transition is underpinned by a need for distributed architectures that can scale to the performance demands of the digital era.

This paper recognizes the transformational shift toward edge computing, where AI and computational processes occur at the data source. It brings into focus the integration of specialized hardware such as Neural Processing Units and accelerated Graphics Processing Units into edge devices. These developments are crucial for overcoming computational bottlenecks and empowering edge devices to support the sophisticated algorithms fundamental to AI applications.

The exploration here draws upon the convergence of hardware advancements, like TPUs designed for deep learning tasks and NPUs offering versatile computational capabilities, with the needs of edge computing. These innovations present a promising landscape for efficiency of AI and sustainability, pointing to a future where computation is not just about speed, but about smart, context-aware analysis and processing done where it matters most.

This research highlights the growing relationship between edge computing and artificial intelligence (AI), promoting a development approach that aligns the quick, effective nature of edge computing with the ambitious growth model once proposed by Moore's Law. The aim is to direct the industry towards a future where AI applications are not only advanced but also mindful of environmental impacts and tuned to real-world needs. By focusing on the integration of new technologies like TPUs and NPUs within edge devices, this study points to a future where computing is as much about intelligent, location-specific processing as it is about speed, ensuring AI operates effectively and responsibly in our everyday environments.

II. MOORE'S LAW AND THE EVOLUTION OF COMPUTING

A. Overview of Moore's Law

Moore's Law, defined by Gordon E. Moore in 1965, predicts the doubling of transistors on a microchip every two years, effectively halving the cost of computers in the process. This principle, foundational to computing's evolution, suggests exponential growth in processing power, guiding the semiconductor industry's strategic planning and research focus [3][4].

Represented mathematically, Moore's Law implies that if $N(t)$ denotes the number of transistors at year t , the count at

year $t + 2$ would be $N(t + 2) = 2N(t)$, highlighting not just growth in computational capabilities but also improvements in efficiency, cost-effectiveness, and device miniaturization.

However, challenges arise as physical and economic barriers to transistor miniaturization become evident, prompting exploration into alternatives like advanced architectures and quantum computing to sustain growth. This law remains significant for understanding shifts in computing trends, particularly as the industry evolves towards specialized hardware to meet modern computational demands.

B. Historical Impacts of Computation development

The evolution of computational technology has been a cornerstone of modern civilization, driving progress in diverse fields ranging from healthcare and engineering to finance and communication. The journey of computational development is marked by landmark innovations, each paving the way for exponential increases in processing power, efficiency, and application domains.

1) **The Dawn of Computing:** The story begins with mechanical computing devices, such as the abacus and Charles Babbage's Difference Engine, which introduced the foundation for programmable machines. These early inventions underscored the potential of mechanized computation but were limited by the technology and materials of their time.

2) **The Electronic Era:** The invention of the electronic computer in the mid-20th century, epitomized by ENIAC (Electronic Numerical Integrator and Computer), marked a pivotal shift [16]. This era ushered in the use of vacuum tubes and later transistors, significantly enhancing computational speed and reliability while reducing size. It set the stage for the development of programming languages and the concept of an operating system, facilitating more complex and varied applications of computer technology.

3) **The Microprocessor Revolution:** The introduction of the microprocessor in the early 1970s democratized computing, making it accessible to businesses and eventually the general public [16]. This period saw the birth of the personal computer, which transformed computing from a tool for scientists and large corporations to an indispensable part of everyday life for millions. The microprocessor revolution also spurred the development of the software industry and the rise of video games, changing entertainment landscapes forever [16].

4) **The Internet and Connectivity:** The advent of the internet in the late 20th century revolutionized information exchange, making it possible to connect computers worldwide into a vast network of data and communication [17]. This global connectivity fostered unprecedented access to information, catalyzed the growth of e-commerce, and enabled the rise of social media, fundamentally altering human interaction and societal structures.

5) **Mobile and Cloud Computing:** The early 21st century witnessed the rise of mobile computing, with smartphones and tablets putting powerful computing capabilities in the palms of users' hands [17]. Simultaneously, cloud computing

emerged, allowing for the storage and processing of data over the internet, which enhanced computational capabilities without the need for advanced hardware in every device. These advances have facilitated the growth of big data analytics, IoT (Internet of Things), and on-demand software services.

6) **AI and Machine Learning Renaissance:** More recently, advances in AI and machine learning have heralded a new era in computational development. Powered by vast datasets and increasingly powerful computational resources, including specialized hardware like GPUs and TPUs, AI applications are reshaping industries, from autonomous vehicles and personalized medicine to intelligent automation and beyond.

7) **Looking Towards Quantum Computing:** As we approach the physical limitations of silicon-based transistors, the computational development landscape is poised for the next revolutionary leap with quantum computing [18]. Offering the potential to process complex problems exponentially faster than classical computers, quantum computing promises to unlock new frontiers in scientific research, cryptography, and material science.

C. The Limitations of Moore's Law in Modern Computation

In today's digital landscape, the exponential growth of data and the widespread presence of Artificial Intelligence (AI) applications have propelled us beyond the confines of Moore's Law. Moore's Law, first created by Gordon Moore in 1965 has served as a guiding principle for the semiconductor industry, predicting a doubling of transistor density on integrated circuits approximately every two years [1]. However, as the demands of modern computing have evolved there has been a rise in physical limitations of semiconductor technology which have become increasingly more apparent as time goes on.

There are several factors contributing to the limitations of Moore's Law in contemporary computation which have been challenging its continued applicability and necessitating alternative approaches to sustain technological progress.

Some potential limitations are that as transistors are miniaturized to ever-smaller dimensions, they approach fundamental physical limits imposed by quantum mechanics. Some issues that arise from quantum mechanics include quantum tunneling, leakage currents, and other quantum effects which have become more pronounced at nanometer scales, leading to increased power consumption, heat dissipation, and reliability issues [5]. The capital expenditures required to develop and maintain the cutting-edge technologies have led to making it economically untenable to sustain the pace of Moore's Law. As a result, the cost of semiconductor manufacturing has skyrocketed especially with the transition to more advanced process nodes, such as below 7nm [3]. The decrease in returns on transistor scaling have led to diminishing improvements in overall system performance, particularly for power-constrained applications.

Researchers are currently exploring alternative methods, materials, device structures, and computing architectures in order to overcome the limitations of traditional semiconductor

scaling. Innovations such as three-dimensional (3D) integration, neuromorphic computing, and photonic computing offer promising avenues for sustaining computational progress beyond the confines of Moore's Law [6].

III. CHALLENGES IN EDGE COMPUTING

While edge computing presents a paradigm shift toward more localized and rapid data processing, it introduces several significant challenges that need careful consideration:

- **Hardware Limitations:** Edge devices vary greatly in their capabilities, which can be a barrier to implementing complex AI algorithms that require substantial computational power. Ensuring that these devices can handle advanced processing tasks without compromising performance is a major challenge.
- **Energy Consumption:** Edge devices often operate on battery power or in environments where energy efficiency is crucial. Developing low-power AI algorithms and hardware that can perform tasks efficiently is critical to the expansion of edge computing.
- **Network Reliability and Bandwidth:** Despite the benefits of processing data locally, edge computing devices often need to communicate with central systems or other edge devices. This reliance on network connectivity means that limited bandwidth and network reliability are potential issues, especially in remote or overloaded networks.
- **Security and Privacy:** With data processing happening on many devices distributed across different locations, securing this data against unauthorized access and ensuring privacy becomes increasingly complex.
- **Management and Scalability:** Deploying and managing a vast number of edge devices, each potentially running different aspects of applications, poses significant logistical challenges. Ensuring these devices can be easily updated and maintained without significant downtimes is crucial.
- **Environmental Impact:** As the number of devices increases, so does the environmental impact associated with manufacturing, deploying, and disposing of these devices. Developing sustainable practices is essential to mitigate this impact.

These challenges underscore the need for innovative solutions in hardware development, software design, and system architecture to fully realize the potential of edge computing in modern AI applications. Addressing these issues is crucial for developing an edge computing infrastructure that is not only efficient and powerful but also scalable, secure, and sustainable.

IV. INNOVATIONS IN COMPUTING HARDWARE

In the unfolding landscape of artificial intelligence (AI), the quest for computing hardware capable of matching the pace of advanced algorithms and expansive data troves is escalating. As we transition beyond the era predicted by Moore's Law, where the compaction of transistors on a chip would double biennially, we encounter a surge in computing hardware innovation. This transformative phase is marked by

a myriad of strategies including Quantum, Neuromorphic, Photonic, and Bio-Inspired Computing, each with distinctive benefits and obstacles in the pursuit to cater to the dynamic demands of AI applications.

Parallel to the exploration of groundbreaking computing architectures, the advent of specialized processing units for AI-centric tasks has become a focal point of innovation in this post-Moore epoch. Amongst these advancements are Google's TPUs, NPUs, and GPUs that have been fine-tuned for AI acceleration. Each of these processors is engineered to expedite particular AI operations, heralding a new era of performance and energy efficiency.

TPUs signify Google's revolutionary stride in expediting machine learning tasks. Tailored for matrix computations prevalent in neural network tasks, TPUs as shown in Fig2 facilitate rapid progress in both the training and deployment stages of machine learning models. They integrate specific hardware enhancements, including fused multiply-accumulate (MAC) units and support for reduced precision arithmetic, to amplify performance while conserving energy.

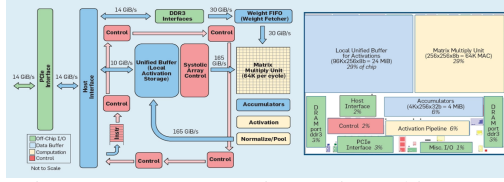


Fig. 2: Tensor Processing Unit Architecture

NPUs as shown in Fig3, or AI accelerators/chips, embody specialized components devised for a spectrum of AI functions extending beyond deep learning frameworks. With their proficiency in performing neural network inference tasks, NPUs enable instantaneous AI processing within edge devices and data centers. Unlike TPUs that are tailored for tensor operations, NPUs offer versatility, adept at handling various neural network structures and algorithms. They feature dedicated instructions and hardware segments geared for standard AI processes such as inference in convolutional neural networks (CNNs), recurrent neural network (RNN) operations, and attention-based mechanisms.

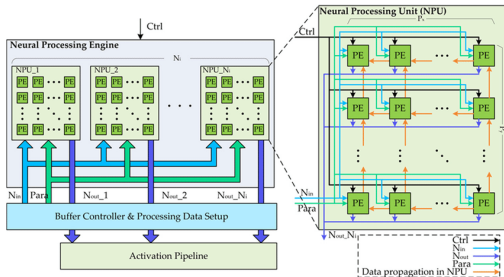


Fig. 3: Neural Processing Engine

GPUs have long been instrumental in fortifying AI operations, leveraging their native parallel structure and computational might. The evolution of GPU technology has introduced specialized cores and software enhancements for AI-specific

tasks, thereby creating AI-accelerated GPUs. These GPUs integrate tensor cores and libraries like NVIDIA's TensorRT and cuDNN to hasten AI operations, including convolutions and matrix multiplications. One of the Nvidia GPU shown in below Fig4.

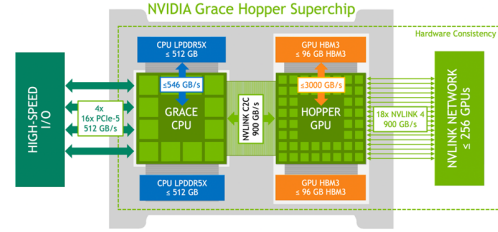


Fig. 4: Graphics Processing Unit

The emergence of specialized processing units such as TPUs, NPUs, and AI-accelerated GPUs signals a transformative shift in the realm of computing hardware design. By delivering exceptional performance and efficiency for AI tasks, they spearhead innovation across both edge and cloud-based AI deployments. The current era of hardware and software symbiotic design is pushing the limits of computing capabilities, propelling us into the post-Moore frontier of technology.

V. ALGORITHMIC ADVANCEMENTS IN AI

In the domain of edge computing, the landscape is rapidly evolving to embrace artificial intelligence (AI) at the edge, where immediacy and efficiency of data processing are paramount [19]. This necessitates a new breed of algorithms tailored for the unique constraints of edge devices—limited computational resources, a premium on energy efficiency, and a need for reduced latency. The advancements discussed here are pivotal, demonstrating an agile approach to AI algorithm development geared towards edge computing environments.

The **Forward-Forward Algorithm** as shown in below Fig5 represents a paradigm shift in neural network training methodologies. Conceived by Geoffrey Hinton, this algorithm deviates from the conventional back-propagation technique by instituting dual forward passes utilizing positive and negative datasets. This facilitates each neural network layer to independently adapt, enhancing the model's 'goodness' metric for accurate data while diminishing it for erroneous inputs. Its inherent efficiency in learning, without reliance on error back-propagation, offers compelling benefits for on-device AI, chiefly the potential for lower power consumption and the facilitation of continuous, real-time data processing [19].

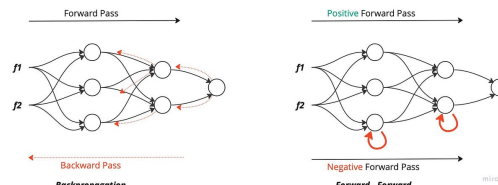


Fig. 5: Forward Forward Algorithm

In the context of computational precision, *Expert-wise Bit Width Adaptation* emerges as a method that dynamically adjusts computational precision across a neural network. By allocating computational resources judiciously—only expending higher precision where it critically impacts performance—this technique optimizes processing efficiency. This is especially relevant for edge AI, where computational overhead must be minimized to conserve battery life and expedite decision-making processes.

In-Memory Expert Management is another strategic enhancement, drawing inspiration from the principles of cache memory. It optimizes on-device memory utilization, ensuring swift data retrieval and processing, much like a cache does in traditional computing systems. This strategy is instrumental in accelerating AI operations on edge devices, enabling them to deliver faster and more efficient data handling capabilities.

Lastly, the *ULEEN Architecture* (Ultra-Low Energy Edge Neural Network) incorporates innovative features such as non-linear thermometer encoding and bloom filtering. This architecture is designed to sharply reduce the energy footprint of neural networks, making it a linchpin for sustainable AI at the edge. Energy conservation is of utmost importance in edge devices, and ULEEN's architecture is specifically engineered to operate within the stringent energy confines of these platforms.

Collectively, these innovations reflect a conscientious push towards the maturation of AI capabilities at the edge. They embody the quest for optimized algorithms that cater to the constraints and performance requirements of edge computing. Through meticulous research and development, these algorithmic enhancements set the stage for AI systems that are not just theoretically adept but are pragmatically deployable in real-world edge scenarios. Such progress is indicative of the AI field's responsiveness to the evolving demands of technology at the periphery of our networks, fortifying the edge with intelligence that is both cognizant of and adaptive to its operating conditions.

VI. CASE STUDIES AND APPLICATIONS

A. Autonomous Vehicle Case Study

In the forefront of the contemporary transportation landscape, the integration of AI within autonomous vehicles heralds a pivotal transition from conventional on-board decision-making to real-time data processing at the edge. Pioneering this revolution, companies like Tesla have leveraged sophisticated AI algorithms and edge computing infrastructure, fundamentally rethinking computing for AI in the post-Moore's law era.

Edge computing is paramount in autonomous vehicular technology, facilitating the swift processing of sensor data and the execution of AI algorithms either directly on the vehicle or proximally at the network edge. This decentralization is crucial for rapid, real-time decision-making, mitigating reliance on distant cloud servers, thus reducing both latency and bandwidth demands. Furthermore, edge computing assures functionality in areas with limited connectivity, leveraging

roadways-infused edge infrastructure and V2I communication networks to deliver essential real-time traffic and condition data to augment safety and efficiency [5][6][7].

Underpinning these advancements are the robust hardware components embedded in these vehicles, from high-performance CPUs and GPUs to AI accelerators. Each generation of hardware delivers exponentially greater computational power necessary for the voluminous and rapid processing of sensor data. Initially, Tesla's utilization of Nvidia's Tegra 3 processor in 2014 marked the inception of this hardware evolution. The processor was equipped with 4 CPUs and 12 GPU cores[8].

By 2016, the technology had advanced to Nvidia's DRIVE PX2 platform, featuring 2 CPUs and 252 GPU cores, delivering an astounding 24 trillion operations per second[9]. Tesla's own Full Self-Driving (FSD) chip as shown in below Fig6, a testament to bespoke innovation, represents the third generation with 6 billion transistors, 12 CPU cores, and 96 GPU cores, capable of performing 36 tera operations per second, catapulting computing performance to a level 21 times that of the second generation[10].

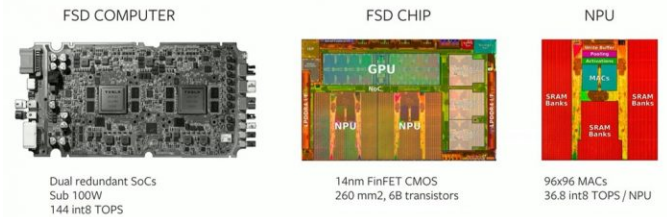


Fig. 6: Tesla chip for Full Self Drive (FSD)

These hardware innovations are more than incremental; they are foundational for autonomous vehicles, enabling the real-time processing required for complex AI algorithm execution. For example, Nvidia's DRIVE platform's contribution to Tesla's capabilities illustrates the computational muscle necessary for cutting-edge autonomous driving applications. This progressive hardware architecture, encompassing custom ASICs and sensor fusion technology, advances the responsiveness and dependability of autonomous driving systems, anchoring future progressions in connected and autonomous vehicular mobility[9][10].

Thus, as we navigate the post-Moore's law era, the automotive industry's pivot to AI-infused edge computing stands as an exemplar of innovation. No longer constrained by transistor density, but by the agility of on-the-edge intelligence, we journey toward an epoch marked by heightened safety and augmented efficiency in transportation.

B. Google Pixel Case Study

The Google Pixel 6 represents a significant stride in the realm of edge computing, driven by its innovative use of on-device Machine Learning (ML) capabilities. These advancements are prominently powered by Google's own mobile

system-on-chip (SoC), the Google Tensor, which combines various processing units to deliver state-of-the-art ML innovations[12].

A pivotal component in the Pixel 6's arsenal is the Neural Architecture Search (NAS) i.e., depicted in below Fig7, which automates the process of creating ML models. This system incentivizes algorithms to find models that not only reach high-quality benchmarks but also comply with constraints like latency and power efficiency. This is particularly critical for the seamless operation of the Pixel 6's camera features, such as real-time face detection that underpins functionalities like focus and exposure adjustments[11][12].

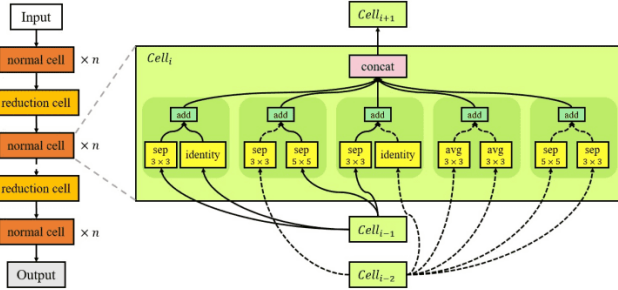


Fig. 7: Neural Architectural Search

At the heart of these on-device vision tasks is the Inverted Bottleneck (IBN) block, a neural network building block that has been adapted for efficient operation on the Tensor TPU. There are various iterations of the IBN, including the depthwise-IBN, fused-IBN, and the group convolution-based IBNs (GC-IBNs) which are shown in Fig8, where each optimized for different stages of the deep learning models to balance computational efficiency and model performance[11].

For the Pixel 6, MobileNetEdgeTPUV2 models, constructed with these IBN variants, deliver superior accuracy for image classification tasks at preferred latencies when executed on the TPU. The model is also optimized for other components in the Tensor SoC, including the CPU, which demonstrates the power of NAS in creating versatile and efficient ML models[11].

In summary, the Google Pixel 6 embodies the latest technological advances in edge computing, offering an on-device ML

experience that's not only fast and efficient but also respects user privacy and power consumption. The use of NAS and the design of specialized hardware, such as the Tensor TPU, underscore Google's commitment to pushing the envelope in smartphone capabilities. These technologies coalesce to redefine mobile photography and on-device intelligence, setting a new benchmark for what smartphones can achieve with edge AI.

VII. CONCLUSION AND FUTURE WORKS

The confluence of edge computing and AI is at a transformative stage, where the vision outlined by Moore's Law is being reshaped by emergent technologies and innovative algorithmic approaches. This case study has traversed the historical trajectory of computation, the burgeoning potential of edge AI, and the bespoke challenges it faces today.

A. Conclusion

- The transition beyond Moore's era heralds an integrated approach to computation, emphasizing bespoke hardware development, sophisticated algorithmic design, and the mitigation of energy consumption as paramount.
- The advancements in hardware such as TPUs, NPUs, and GPUs specifically optimized for AI applications have heralded a new dawn for edge computing—ushering in an era where deep learning tasks are handled with remarkable speed and efficiency.
- Edge computing's ascendancy has been catalyzed by the symbiosis between these hardware advancements and breakthroughs in AI algorithms. Such as the forward-forward algorithm, offering alternative learning paradigms for neural network training, which are well-suited for the low-power, high-efficiency demands of edge devices.

B. Future Works

- Investigative efforts should intensify around algorithmic advancements such as the forward-forward algorithm, which may simplify the learning process for neural networks on edge devices due to its non-reliance on error backpropagation.
- ULEEN's architecture, with its promise of ultra-low energy neural networks, and methodologies like expertwise bitwidth adaptation, which fine-tunes computational precision, should be examined for their potential to conserve energy while maintaining high AI performance.
- The role of in-memory expert management in leveraging existing memory infrastructure for more efficient data processing, analogous to how cache memory is used in traditional computing systems, will also be a crucial area of focus.
- Continued research into the deployment of TPUs, NPUs, and GPUs designed for AI within edge devices will remain central. These components are essential for real-time AI tasks such as image processing, natural language

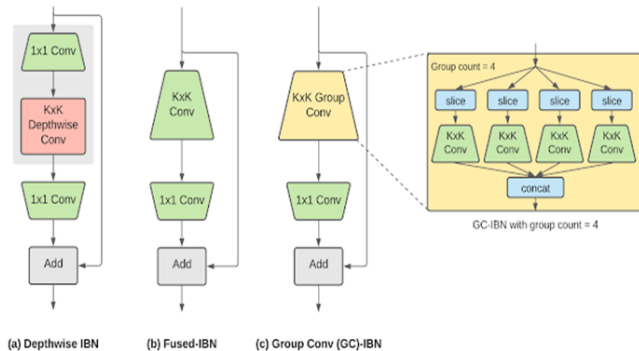


Fig. 8: Group Convolution Based IBN

processing, and autonomous vehicle navigation, where immediate data processing is critical.

- The adoption of group convolution-based inverted bottlenecks in neural network design, particularly for vision-based modeling, is another avenue that shows great promise in balancing computational resources and model efficacy.
- A balanced and forward-thinking approach to computational development is paramount, harmonizing the immediacy and efficiency of edge computing with sustainable and responsible innovation. This includes addressing the environmental implications of increased computational demands.

The future work in edge computing and AI requires a multifaceted strategy encompassing a vast array of disciplines and areas of expertise. It demands a collaborative effort that spans industries, academia, and research institutions to foster innovations that will not only push the computational envelope but also ensure that such advances are sustainable and ethical. As we look to the horizon, it is clear that the legacy of Moore's Law in inspiring computational growth will continue in a new form—one that is acutely attuned to the realities of modern technology and the needs of our global ecosystem.

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